

Week 5 - Task 15: Incident Detection — Investigate an Attack in Logs

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1. Objective

Detect and investigate a brute-force/suspicious login attack using Windows Security logs in Splunk: find the failed-login spike (4625), identify source + attempt count, check for a related success (4624), and build an attack timeline.

2. Tool / Environment

- **SIEM:** Splunk Enterprise
- **Log source:** WinEventLog:Security
- **Host under investigation:** KALI

3. Failed Logon Detection (Event ID 4625)

Query: source="WinEventLog:*" EventCode=4625

Field	Value
Total failed logon events	31
Account targeted	sshtest
Domain	WORKGROUP
Computer	KALI
Logon Type	8 (NetworkCleartext — used by services like SSH)
Caller Process	C:\Windows\System32\OpenSSH\sshd.exe
Failure Reason	Unknown user name or bad password
Keywords	Audit Failure

Finding: All 31 failed attempts targeted the same non-existent/invalid account (sshtest) via the OpenSSH service on host KALI — a classic brute-force/password-guessing pattern against SSH.

4. Source Analysis

Query: source="WinEventLog:*" EventCode=4625 | stats count by Source_Network_Address

Source Network Address	Count
(blank/not logged)	19
127.0.0.1 (localhost)	3

Finding: The attack originated from the same machine (localhost, 127.0.0.1) — consistent with a controlled lab/simulation environment where the attack tool was run locally against the SSH service. 19

events did not log a source address field (common when the field isn't populated by the logon type/process).

5. Successful Logon Check (Event ID 4624)

Query: source="WinEventLog:*" EventCode=4624

- Total 4624 events on the host: **2,244** (normal system/service activity, unrelated baseline volume)
- Sampled 4624 events show Account_Name=SYSTEM, Account_Domain=NT AUTHORITY — standard OS/service logons, **not** the sstest account used in the attack.

Finding: No successful logon (4624) was found for the sstest account or matching the attack's source/timeframe. The brute-force attempt **failed** — attacker did not gain access.

6. Attack Timeline

Time (PM)	Event
12:19:30.882	Failed logon attempt #1 (4625) — account sstest
12:19:42.738	Failed logon attempt #2 (4625) — account sstest
12:19:51.945	Failed logon attempt #3 (4625) — account sstest
...	Pattern repeats — 31 total failed attempts in rapid succession
—	No 4624 success event ever recorded for this account

Attempts occurred within seconds of each other — automated/scripted brute-force behavior, not manual human login attempts.

7. Verdict

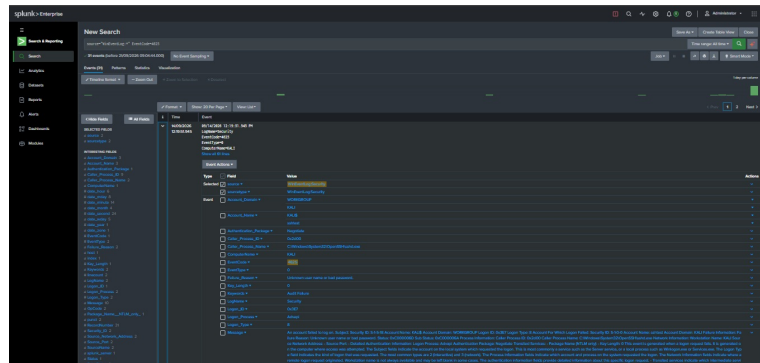
- **Attack type:** SSH Brute-Force / Password-Guessing Attack
- **Target account:** sstest
- **Source:** Localhost (127.0.0.1) — lab-simulated attacker
- **Total attempts:** 31
- **Success:** No — all 31 attempts failed
- **Recommendation:** Lock/disable unused sstest account, enable account lockout after N failed attempts, monitor OpenSSH logon failures, and alert on high-frequency 4625 events from a single process/source.

8. TryHackMe Room Completed (Proof)

Room: Investigating Windows — **100% completed**

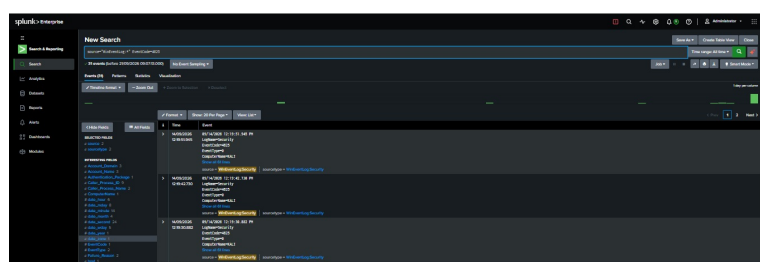
9. Screenshots

Splunk — 4625 Failed Logon Search (Event Detail)



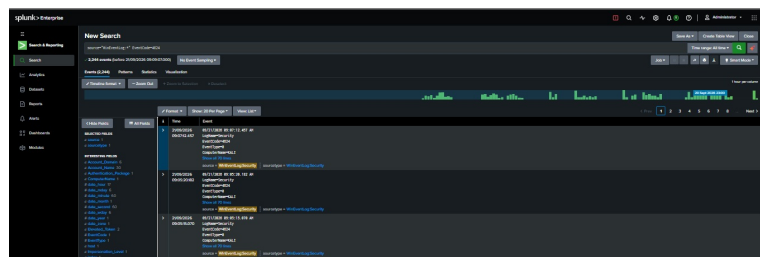
Failed Logon Detail

Splunk — 4625 Event List (Timeline of Attempts)



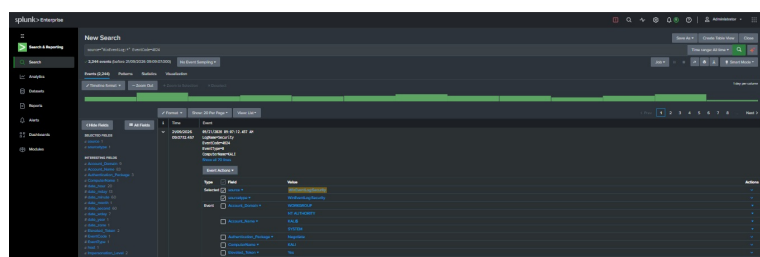
Failed Logon Timeline

Splunk — 4624 Successful Logon Search (2,244 events)



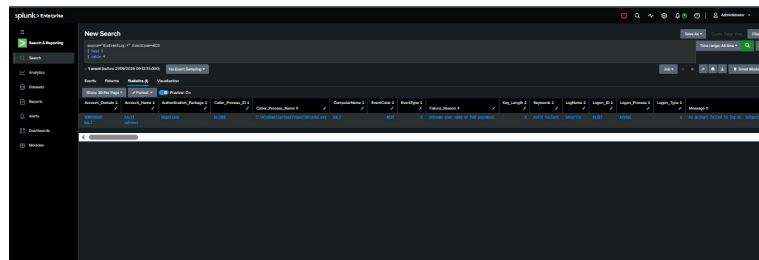
Successful Logon Search

Splunk — 4624 Event Detail (SYSTEM account, unrelated to attack)



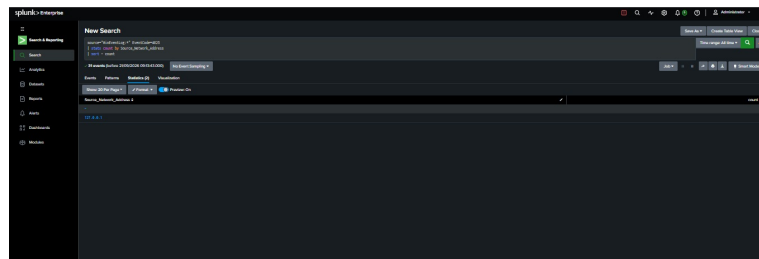
Successful Logon Detail

Splunk — 4625 Event Table (Account, Process, Failure Reason)



Failed Logon Table

Splunk — Source Network Address Breakdown



Source Address Stats

TryHackMe — Investigating Windows (Room Completed 100%)



TryHackMe Room Completed

End of report.